PharmaColdOps Project Report

IRS Practice Module | Project Group 41 | 17 October 2026

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Executive summary

PharmaColdOps is an explainable decision-support prototype for pharmaceutical cold-chain temperature excursions. We connect simulated temperature observations, rule-based disposition recommendations, human review, replacement delivery planning, and case-specific evidence queries in one runnable system. The prototype covers the IRS technique groups of decision automation, knowledge discovery and data mining, resource optimisation, and cognitive systems.

The principal result is an implemented, auditable workflow rather than a validated medicine-quality predictor. M3 agreed with historical adjudicated labels in 54 of 57 synthetic scenarios, but missed all three historical quarantine cases. The shipment-risk experiment achieved five-fold PR-AUC 0.7087 and F1 0.6635; monthly candidate-cause classification remained weak. A subsequent event-level simulation study improved held-out macro-F1, while retaining substantial multi-fault and extreme-condition failures. Static routing experiments produced complete feasible solutions in 49 comparisons. Independent M6 evaluation remains incomplete: reviewer A supplied both annotation files, while reviewer B's files and any necessary adjudication have not been collected. We report that gap explicitly.

Operational orders and event observations are simulated. Engineering temperature limits are not manufacturer stability limits, model candidates cannot authorise disposition, and human review governs the effective result used by downstream delivery. The deployment is a local demonstration with persistent storage and recovery checks, not a public or clinically validated service.

1 Business problem and intended value

An excursion creates two connected work requirements. Quality staff must reconstruct the exposure and justify a disposition, while dispatchers must maintain supply if replacement is required. A temperature alert alone does not connect those records. Our project investigates whether explicit rule paths, stable case identifiers and delivery constraints can make this process easier to inspect and demonstrate.

The intended users are quality reviewers, dispatchers, warehouse managers and auditors. The prototype lets them inspect the original observations, distinguish an automatic recommendation from a human decision, compare logistics alternatives and retrieve the evidence associated with the recorded case. Expected benefits include reduced duplicate data entry and clearer handover. We did not conduct an operational controlled trial or measure realised cost savings, medicine wastage reduction or staff productivity.

Commercial monitoring and quality platforms already provide shipment visibility and may support stability-profile workflows. We therefore do not claim that competitors merely raise alarms or that this project invents automated disposition. The differentiating educational contribution is an inspectable integration of rules, ML experiments, constrained routing and graph evidence. The market discussion in the submitted proposal is contextual research, not a current vendor benchmark [1].

2 Domain foundations and method selection

WHO TRS 961 Annex 9, EU GDP, ICH guidance and public SOP references provide principles for storage, transport, investigation and stability evidence. We distinguish those principles from our

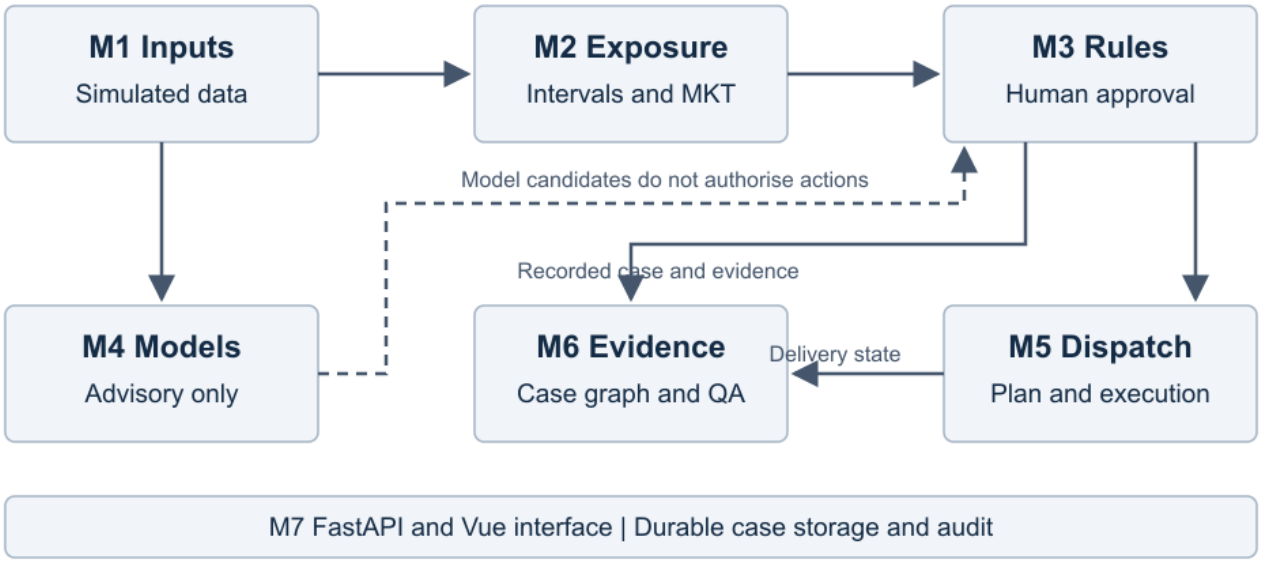
numerical engineering assumptions. A citation to ICH or WHO does not validate a configured excursion duration or mean kinetic temperature ceiling for a specific product [2].

We use deterministic priority rules where an operator must inspect the decision path. The rule engine has release, retest and scrap outputs under its current policy; quarantine remains a historical evaluation label and an available human outcome. Reshipment is a separate logistics action, not a fifth mutually exclusive disposition. Boundary, missing-evidence and low-temperature cases can require human review without rewriting the automatic snapshot [3].

For routing, Solomon VRPTW instances provide standard capacity and time-window problems. Greedy construction supplies an interpretable baseline; OR-Tools RoutingModel with Guided Local Search supplies time-bounded search; a genetic algorithm provides an additional experimental comparison. This is Routing Solver, not CP-SAT, and time-limited results do not prove optimality [6].

For tabular ML, logistic regression and histogram gradient boosting provide reproducible baselines. Offline SHAP and calibration experiments examine behaviour without treating associations as causes. For question answering, fixed intent routing and template Cypher queries allow explicit evidence and failure states. M6 is not an LLM or unrestricted semantic question-answering service [4, 7].

3 System architecture



Implemented module flow

The seven modules share case, order and product identifiers while preserving different responsibilities.

Module	Implemented responsibility	Principal output
M1	Dataset validation and reproducible simulated business inputs	Validated features and seeded order batches
M2	Explicit temperature intervals, hot/cold windows and duration-weighted MKT	Exposure summary with coverage and source observations
M3	Priority rules, evidence mappings and review requirements	Original recommendation and rule path

Module	Implemented responsibility	Principal output
M4	Risk and candidate-cause experiments and advisory inference	Versioned scores and candidate snapshots
M5	Constrained delivery planning and execution	Plans, operation state and explicit infeasibility reasons
M6	Neo4j case chains and structured evidence queries	Case-specific answers and provenance
M7	FastAPI, Vue, SQLite persistence and Docker integration	Bilingual online interface and offline rule sandbox

The main path is observation to assessment, registration, human review when required, replacement demand, route selection, execution and case closure. The advisory model path does not override M3. An accepted human outcome is stored separately and becomes the effective disposition for subsequent workflow checks. Historical rule and model snapshots remain unchanged [3, 8].

The API commits registration and its graph-outbox entry in one SQLite transaction. A background worker leases pending entries, retries failures and replays complete case chains idempotently into Neo4j. An outbox entry marked synced is not treated as proof that every graph relationship still exists: the audit endpoint can compare actual graph coverage to authoritative case records [8].

The Vue client supports same-origin API mode and an explicit offline rule sandbox. Offline mode does not register cases, write the graph or execute deliveries. The map displays cached road geometry and simulated progress rather than GPS or live traffic. The current operational deployment uses one API worker and loopback binding; distributed logistics transactions and authenticated multi-user operation are outside the demonstrated architecture [9].

4 Implementation and workflow

4.1 Temperature assessment and human review

M2 consumes explicit simulated temperature intervals and records known coverage. It calculates duration-weighted MKT, contiguous excursion windows and the selected event range. Missing intervals are not replaced with fabricated readings. A wholly unassessed sequence does not acquire an invented scalar temperature or MKT. The API recalculates derived values rather than trusting a mismatching client summary [10].

M3 records product, stage, packaging, exposure, rule number, reason and mapped evidence. Registration is durable and idempotent by draft identity. A boundary result can be registered for review without being treated as final approval. Pending review blocks linked downstream actions; an explicit human decision records reviewer identity, reason and time. Version checks reject stale mutations, and executed outcomes cannot be retroactively changed [3].

Case audit export provides printable HTML and a full JSON snapshot. It includes original assessment, human review history, recorded model identity, associated delivery state and actual graph coverage. It does not recompute an old case with a newer model. A content hash detects a changed snapshot but is not a cryptographic signature or compliance certificate [11].

4.2 Delivery planning and execution

The order generator records effective configuration, date, seed and source hashes. Operators can demonstrate routine demand, tight delivery windows, multiple sources or temperature zones, and insufficient resources. Real hospital orders, exact cargo quantities and warehouse stock balances are not required for this teaching scenario [12].

Independent urgent orders use the same operation and candidate constraints as ordinary replacement demand. Plan previews do not reserve a vehicle; adoption is an explicit action with version and retry protection. Execution uses a simulated clock. The final allowed parking facilities are a user-selected subset of five warehouses; hospitals cannot be day-end terminals. Parking must complete before a new day inherits vehicle positions [12].

Delay inspection identifies late remaining work and can preview reordering under the same constraints. Mechanical failures freeze the affected vehicle and retain failed original orders while separately tracking feasible replacement work. The demonstration does not invent vehicle-to-vehicle transfer, towing or repair capability. PDPTW in-transit rescue is explicitly unsupported rather than silently approximated [13].

4.3 Advisory models and graph queries

Risk and monthly candidate-cause models require explicit input contexts and save their first assessment with model identity. The independent event-v2 interface uses 111 allowed observation-derived features and a trusted read-only model bundle. Candidates and abstentions both require review; automatic actions remain disabled. Invalid or unavailable bundles produce an explicit unavailable state rather than a substituted model [5].

M6 supports disposition reasons, audit chains, product requirements, cause context and disposition statistics. Missing cases, insufficient evidence and graph unavailability are distinct states. The frontend's keyword router may fall back to a conceptual FAQ, which is not counted as a successful live graph query merely because it displays an answer [7].

5 Data and evaluation design

There is no public dataset in this project with independently validated medicine release, quarantine, retest and scrap decisions. We therefore evaluate each component against an appropriate and explicitly limited reference. The M3 scenario bank contains 57 synthetic scenarios with historical human adjudication. ML sources and licences are documented separately; the shipment-risk dataset is suspected synthetic and the monthly cause dataset is explicitly simulated [2, 4].

The risk task uses 8,000 shipment rows, including 1,574 failures, with 13 summary features. The cause task filters 16,192 excursion-month rows into ten candidate-cause classes with 15 features. The latter has 447 provided facility-ID groups and 48 months, but these are not verified real institutions or event timestamps [4].

The formal ML protocol separates fitting, calibration, selection and final outer testing. Risk uses five stratified outer folds because its source lacks timestamps. Cause experiments include random, provided-facility-ID and month-forward folds, with full and context-only feature sets. Preprocessing and weighting use the fitting partition; test data do not choose thresholds. These are separate tasks from event-v2 simulation and must not share an accuracy headline [4].

Event-v2 has 10,200 simulated events and 9,597 families across training, validation and five held-out roles. Models and screening policy were frozen before final test generation. Asset, family and excitation

controls reduce shortcut leakage, but all cohorts still come from the same uncalibrated simulator. They do not establish external physical or clinical validity [5].

6 Experimental results

6.1 Rule engine

M3 agreed with 54 of 57 historical labels, or 94.7368%. Macro-F1 was 0.717437. Engine-to-gold Cohen's kappa was 0.909667; historical annotator-to-annotator kappa was 0.6434. These are different statistics. All three quarantine-labelled cases were missed because the current frozen engine policy has no automatic quarantine output. We retain those cases and the four-class denominator [3].

Historical class	Support	Recall	F1
release	13	1.0000	0.9286
retest	8	1.0000	0.9412
quarantine	3	0.0000	0.0000
scrap	33	1.0000	1.0000

The three discrepancies are S034, S035 and S052. They represent a difference between the written policy and historical adjudication, not proof that either side is clinically correct. Human review protects the operational workflow but does not improve this frozen automatic-engine score [3].

6.2 Shipment risk and monthly candidate causes

Risk metric	Five-fold result
F1 at selection-chosen thresholds	0.6635 +/- 0.0179
ROC-AUC	0.9045 +/- 0.0092
PR-AUC	0.7087 +/- 0.0296
Brier score before and after calibration	0.1091 to 0.0933

For full15 monthly cause classification, macro-F1 was 0.1322 under random folds, 0.1308 under provided-ID folds and 0.1259 under month-forward testing. Corresponding Top-1 results were 0.1713, 0.1500 and 0.1497. Calibration improved probability losses but did not change class rankings. Weak extra feature signal, contextual shortcuts and insufficient event-level information motivated the separate learnability audit and simulator work [4].

Offline Kernel SHAP was matched to the relevant model identities and checked for additive reconstruction within its numerical tolerance. Online reference-feature replacement remains a sensitivity display and is not relabelled as SHAP. Neither method proves the cause of an individual event [4].

6.3 Event-level temporal study

Held-out role	Selected temporal v2 macro-F1	Frozen earlier event model macro-F1	Supported candidate coverage
nominal new assets	0.7668	0.6125	17.33%
other new assets	0.7619	0.6111	20.67%
unseen excitation	0.7228	0.5880	13.83%
covered stress	0.7528	0.5126	2.50%
more extreme stress	0.6505	0.5068	0%

Among supported candidates, complete-set precision was 96/104, 116/124 and 75/83 in the first three roles. Covered stress had only 15 supported events; 15/15 does not establish a population guarantee. Extreme stress supported none, so precision is unavailable rather than 100%. All 3,000 test events still required human review [5].

Raw normal-event false alarms remained 47.73-90.80%. Extreme multi-fault complete-set accuracy fell from 11.96% to 8.70% despite improved recall. Thermal-buffer F1 remained approximately 0.079-0.260. Observationally identical door/staff and power/fuel mechanism combinations show why tuning alone cannot resolve every cause label. The system must expose ambiguity and request additional evidence [5].

6.4 Static routing

The formal comparison has 49 solver runs across six Solomon instances and one Singapore instance. All runs served every target without recorded hard violations. On Singapore, greedy used three vehicles and 165.26 km; OR-Tools used three and 137.85 km. This is a 16.59% distance reduction for that frozen static instance, not an operational savings estimate [6].

OR-Tools matched the published two-decimal reference distances on C101 and C201. R101 and R201 had same-fleet distance gaps of approximately 0.1656% and 0.1753%. RC101 used 16 vehicles against the reference 14, so its shorter distance cannot be described as beating that reference. GA did not outperform OR-Tools in this experiment. Each configuration had one run; no multi-seed significance claim is made [6].

These are static single-origin algorithm comparisons. Urgent insertion, grouped operations, delay recovery, failure handling and overnight parking are implemented workflow features verified separately, not additional measured rows in that benchmark [6, 13].

6.5 Knowledge graph and independent human evaluation

The frozen M6 preparation performed 12 actual service queries and 41 client conceptual fallbacks on 53 questions. The same 57 case scenarios passed 466 derived contract checks for record agreement, mapped references, thresholds, states and case isolation. Derived checks do not establish independent answer correctness or citation relevance [7].

Reviewer A has supplied 53 intent labels and 53 answer/evidence ratings in separate completed files. The project owner confirmed human authorship on 2026-10-17. Structural checks verify IDs, allowed values and frozen hashes. Reviewer A assigned correctness 2 to 28 answers, 1 to 21 and 0 to four;

evidence relevance was 0 for 41, 2 for eight and not applicable for four. Unsupported claims were flagged in ten answers. These are one reviewer's observations, not dual-rater adequacy or independent system accuracy.

Reviewer B's two files are unavailable in the repository. Human-human kappa, adjudicated intent accuracy and the proportion rated adequate by both reviewers are therefore unavailable. The A-rating upload also records corrections to reviewer ID and source URLs; actual consultation of replacement sources must be documented by the reviewer. Source strings and valid hashes cannot establish provenance, independence or professional competence [7].

7 Reliability and reproduction

The 2026-10-17 isolated regression passed 909 tests with zero skips and six existing warnings; a separate API walkthrough passed 36 checks against its own test graph. Frontend production build and eight frontend check scripts passed. Twenty live advisory sample replays and two browser preview branches preserved the original case records. These are software checks, not new model-accuracy observations [9].

An initial run passed 892 tests and failed one because the temporary QA fixture had not loaded road relationships. We preserved that result, loaded the complete standard static seed in a new test graph and reran successfully. The historical 2026-10-04 baseline of 888 tests remains a separate record. Local map-tile and no-active-operation 404 responses, plus a large-JavaScript-chunk warning, remain disclosed presentation limitations [9].

The deployment pins Python, frontend and Docker dependencies, runs the API as a non-root user with read-only root filesystem, and persists authoritative case data independently of Neo4j. Previous acceptance exercised actual API restart, graph outage, outbox recovery and backup restoration into new volumes. These checks demonstrate recovery for the tested configuration, not high availability or a production service-level agreement [8, 9].

For reproduction, clone the repository, create the documented Python 3.12 environment, install the evaluation lock, and use the Linux-container Docker path. Native Windows workflow mutations are unsupported because their file guard requires Unix fcntl; the standalone M6 format checker is portable. Native Linux amd64 and arbitrary offline builds have not been established by the historical arm64 checks [9].

8 Findings and remaining work

The implemented rule, review, logistics and graph boundaries make failures inspectable. In particular, a reviewer can trace original evidence without conflating it with a later human outcome, and a failed graph write does not silently erase the authoritative case. The experimental results also limit the product claim: historical policy disagreement, weak monthly causes, multi-fault omissions and incomplete M6 human evidence prevent a claim of autonomous pharmaceutical judgement.

Before submission, the team must complete or explicitly disclose the missing M6 independent review, document source verification and seal times, record two five-minute videos with human speech, and submit individual reflections and peer evaluation privately. A frozen source package and a report are not evidence that these human deliverables exist. The delivery checklist records the remaining inputs.

References and project evidence

1. Submitted proposal, PharmaColdOps-Proposal-Group 41.pdf, 2026-09-13. Reproduced as Appendix C in the PDF report.

2. Threshold evidence audit and data dictionary. Public anchors: WHO TRS 961 Annex 9 and EU GDP. The project audit records applicability and assumptions.

3. M3 formal evaluation and human effective dispositions. Frozen formal evaluation, 2026-10-03.

4. M4 formal experiments and cause-data learnability audit. Formal folds, calibration, ablations and model-matched explanations.

5. Temporal event-v2 evaluation and shadow API and UI. Independent event study and advisory contract.

6. M5 formal evaluation. Solomon, M. M. (1987), Algorithms for the Vehicle Routing and Scheduling Problems with Time Window Constraints, original paper; frozen comparison documented against SINTEF reference results.

7. M6 independent evaluation, reviewer A intent labels and reviewer A answer ratings. Individual review inputs, not a complete two-rater report.

8. Graph reliability and recovery and case audit export. Authoritative storage, graph coverage and recovery scope.

9. Reproduction and deployment, controlled v2 deployment and progress log. Versioned deployment and historical verification records.

10. M2 temperature and MKT. Interval semantics, MKT and missingness contracts.

11. Case audit export. Snapshot and hash scope.

12. Simulated order generator and urgent orders and parking. Simulated business-input and parking constraints.

13. End-to-end acceptance and routing design. Dynamic workflow scope and acceptance protocol.

Appendix A Mapping to modular course techniques

Course area	Applied knowledge and implemented functionality	Evaluation evidence
MR Machine Reasoning	Knowledge/model representation, reasoning as search and inductive learning: explicit rule paths, constrained route construction and tabular risk/cause models	57 rule scenarios; search baselines; five-fold ML and held-out simulations
RS Reasoning Systems	Optimisation and evolutionary computation: VRPTW formulation, RoutingModel GLS and permutation-based GA; graph-based system integration	Static 49-run routing comparison and separate dynamic execution acceptance
CGS Cognitive Systems	Knowledge representation and reasoning with a case/rule/evidence graph and structured user questions; five template queries with explicit failure states	Derived contract checks; independent M6 review still pending; no audio or vision claim

This mapping follows the course agendas in the IRS practice-module briefing, page 3. The separate four technique groups are decision automation (M3), business resource optimisation (M5), knowledge discovery/data mining (M2/M4), and cognitive systems (M6). It does not assert that every optional feature or course topic, such as audio/vision cognition, was implemented.

Appendix B Installation and user guide

B1 Fresh demonstration deployment

Requirements are Docker Desktop in Linux-container mode, a free loopback port 8080 and Python 3.12. Create an environment with python3.12 -m venv .venv, then install requirements-eval.lock.txt. On Windows use .venv/Scripts/python.exe in place of .venv/bin/python. First builds require registry, PyPI and npm access.

```
.venv/bin/python scripts/configure_demo.py
docker compose --env-file deploy/demo.env -f deploy/compose.demo.yml config --quiet
docker compose --env-file deploy/demo.env -f deploy/compose.demo.yml up -d --build --wait --wait-timeout 180
```

Open http://127.0.0.1:8080/?lang=en and check /api/ready. This base deployment defaults to event-v2 disabled. For the existing v2 demonstration use both env files and both compose files from the controlled deployment guide; do not recreate that deployment using only the base configuration. Never add -v when stopping the application's persistent volumes.

B2 Operator walkthrough

Generate a seeded daily batch and inspect the date, time windows, fleet and allowed final warehouses. Preview the plan and confirm departure. Pause simulated progress while demonstrating an urgent order or registering an excursion linked to the correct original order and occurrence facility. Examine the original rule recommendation and its evidence. Complete human review if required, compare feasible replacement candidates, then adopt one explicitly.

Resume simulated progress and inspect actual recorded delivery status. Complete case notes only after required delivery conditions are met. Ask for the selected case's disposition reason or audit chain; distinguish a conceptual FAQ from graph evidence. Export its HTML/JSON audit. At day end, complete legal warehouse parking before starting the next day. Hospitals are not final parking sites.

B3 Failure handling and offline mode

Readiness failure can indicate unavailable storage, models or graph dependencies. Use the documented compose ps/logs commands for the actual selected deployment. A map without online tiles can still display the route geometry. Offline rule mode remains available for presentation, but must be disclosed as offline and cannot execute deliveries. Do not run a graph loader against a populated deployment to fix a missing relationship; use the audit and non-conflicting repair procedure.

The full English manual and Chinese manual provide panel-level instructions, troubleshooting and the separate M6 review process. This appendix summarises installation and operation.

Appendix C Submitted project proposal

The PDF report appends the unchanged 2026-09-13 submitted proposal. It is a historical planning document; current implementation, experiment results and outstanding work are described in this report. The editable DOCX contains this appendix introduction and identifies the original PDF as a companion source.

The historical proposal cover still says Project Group 52. The revised proposal source and current presentation identify the team as Project Group 41, which is used on this report and member-github.txt. The four member identities are unchanged. We retain the submitted appendix as a historical record rather than silently alter its cover.

PharmaColdOps

Decision Support for Cold Chain Pharmaceutical Excursions and Delivery Replanning

Course: IRS Practice Module

Group name: Project Group 52

Members: Xu Wenzhe (A0328771W), Zhu Jianyu (A0353769L), Wang Lepeng (A0357864L), Shen Ziyi (A0350940J).

1. Project Overview

PharmaColdOps supports temperature-excursion assessment in pharmaceutical transport and storage. A deterministic rule engine produces traceable disposition recommendations, with a delivery optimiser and knowledge graph intended to connect replacement demand to routes and evidence. Batch dispositions are release, quarantine, retest and scrap. Reshipment is a separate logistics action, not a fifth mutually exclusive disposition.

The project covers four IRS technique groups: decision automation, resource optimisation, knowledge discovery and data mining, and cognitive systems. The delivery objective is a minimal integrated workflow in a controlled Singapore scenario: temperature event → disposition recommendation → reshipment order → delivery plan → evidence query. Risk and candidate-cause classification remain offline experiments and do not enter the disposition API.

By now, the rule engine, bilingual interface, independently annotated scenarios, offline ML baselines, Singapore routing, order-driven reshipment solving and KG case-write code have been developed. The frontend calls the live routing and graph QA endpoints for archived reshipment cases. An order-driven dispatch screen and an expanded destination pool remain to be built. This prototype does not replace final quality approval or claim production or regulatory validation.

2. Problem Definition

After an excursion, quality staff must check product requirements, exposure conditions and packaging, record their reasoning, and coordinate continued supply. The project investigates three testable questions:

- Can written rules be implemented consistently while exposing differences between the specification and human judgement?
- Under vehicle-capacity and delivery-window constraints, how much can optimisation improve distance and service feasibility over a greedy baseline?
- Can a user trace a closed case to its inputs, rule path and evidence, with explicit feedback when records are missing?

Reduced assessment time, fewer inappropriate dispositions and less supply disruption are intended benefits. There is no operational controlled trial, so these are not reported as achieved, quantified benefits.

3. Background and Domain Foundations

3.1 Regulatory Principles and Applicability

Sources include WHO TRS 961 Annex 9, EU GDP 2013/C 343/01, ICH stability and quality-risk guidance, and public CDC/WHO procedures. Acceptable excursion windows for an individual product still require applicable stability evidence. WHO (https://www.who.int/publications/m/item/trs961-annex9-modelguidanceforstoragetransport), EU GDP (https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32013C0343), ICH (https://www.ich.org/page/quality-guidelines)

The project distinguishes public domain principles, product stability evidence and engineering assumptions. Configuration records sources and limitations, but allowable durations, MKT ceilings, some retest settings and rule multipliers include engineering assumptions. These are not universal numerical limits prescribed by regulation. Linking a regulatory node does not validate a particular product threshold.

CDC excursion guidance calls for holding affected vaccines from use and seeking advice from the immunisation programme or manufacturer. Prototype outputs are recommendations for quality review, not authorisation to release or destroy actual products using demonstration thresholds. CDC storage and handling guidance (https://www.cdc.gov/pinkbook/hcp/table-of-contents/chapter-5-vaccine-storage-and-handling.html)

3.2 Related Work and Method Selection

Solomon's VRPTW benchmark supplies standard routing problems with capacity and time-window constraints. The project compares algorithms on six instances and uses Singapore road matrices for the geographic demonstration. OR-Tools Routing Solver provides route construction and local search; the implementation uses time-bounded Guided Local Search. Solomon, 1987 (https://doi.org/10.1287/opre.35.2.254), OR-Tools (https://developers.google.com/optimization/routing/routing_options)

Risk experiments compare logistic regression, LightGBM and XGBoost, with SHAP explaining dependence on tabular features. The cognitive component uses structured graph queries over stored cases, rules and evidence. Original method papers are listed in §14. Feature attribution and cause-label classification on observational data do not establish causality.

The contribution is integration and reproducible evaluation in a controlled workflow, without claiming a new foundational algorithm or the first cold-chain use of these techniques.

3.3 Market Context and Positioning

Commercial products already cover monitoring, shipment visibility and some quality-process automation. For example, Controlant describes Product Stability Automation that determines release status from product stability profiles. It is therefore inaccurate to describe existing vendors collectively as alert-only systems without disposition capabilities. Controlant applications (https://www.controlant.com/applications)

PharmaColdOps is positioned as a teaching and research prototype with inspectable rules, reproducible experiments and demonstrable delivery integration. Comparison dimensions include rule transparency, evidence tracing, routing interfaces and deployment scope. Unverified competitor capabilities remain unknown; missing public documentation is not evidence of absence. Market value still needs user interviews and operational validation.

Decision support capabilities and project scope

Verified example and research scope as of 2026-09-12

Capability	Commercial example	PharmaColdOps
Quality release	Controlant describes stability profile automation	Configurable rule prototype with explicit assumptions
Delivery planning	Not assessed in this review	Greedy and Routing Solver Event integration pending
Evidence	Vendor capabilities require product-specific verification	Stored case and source links Frontend queries pending
Validation	Not benchmarked here	57 human-labelled cases Public data and simulated orders

Source: controlant.com/applications. Unknown capabilities are not treated as absent.

This is a scope comparison, not a market ranking or proof of commercial superiority.

Figure 1: Cold-chain decision-support capabilities and project scope, not a vendor performance ranking.

4. Stakeholders and Intended Value

User	Work requirement	Prototype support
Quality lead	Review excursions and disposition evidence	Input snapshots, rule paths, evidence sources and limitations
Dispatcher	Plan deliveries following replacement demand	Candidate routes under capacity and time-window constraints
Warehouse and transport manager	Investigate factors associated with exceptions	Offline risk analysis and candidate causes
Auditor	Reconstruct an assessment	Case records, linked identifiers and evidence queries

People retain responsibility for final approval and delivery execution. The project collects no patient data and evaluates no clinical outcomes.

5. Overall System Design

5.1 Target Workflow

1. Enter an event with a known product category. Current inputs are forms or demonstration data; sequence parsing is a subsequent data task.
2. Return one of four dispositions, a reshipment flag, rule number, reason and evidence states.
3. Have quality staff review the recommendation. Current closure stores inputs and the system recommendation; fields for final human decisions and approvers remain to be designed, while dispatch run state separately records acceptance, departure and per-order delivery.

4. For replacement cases, submit a simulated order with destination, quantity and time window under the agreed contract and solve it with the single-depot solver. The contract, solver and run persistence are implemented and tested; the frontend has no dispatch screen yet.
5. Link the case and route, display stops, distance, schedules and unserved orders, and query case-specific graph evidence. Archived reshipment cases already fetch live routes and case-specific graph answers; a screen for choosing destinations and quantities remains to be built.

Archiving a case does not establish its real-world outcome. Automatic feedback learning is outside the MVP.

5.2 Current Progress

Module	Status as of 2026-09-13	Remaining work
M1 Data preparation	Data, dictionary, licence checks and audit scripts available	Task definitions and reproducible environment
M2 Event generation	Demo export available with single-reading MKT approximation	Sequence windows, MKT implementation and validation
M3 Rule engine	Four category prototypes, six rules, API, human gold for 57 cases	Domain review, disagreement analysis and formal evaluation
M4 Risk and causes	LR/LGBM/XGB, SHAP and ten-class cause experiment available	Multi-fold evaluation, feature availability and failure analysis
M5 Routing	Greedy solver, Routing Solver, real road network, reshipment/dispatch solving and run persistence available	Order-driven dispatch screen and expanded destination pool
M6 Graph and QA	Schema, loading, closure writing, query functions, HTTP endpoints and frontend answers available	Evidence evaluation and relevance scoring
M7 Integration	Bilingual Vue, offline/API modes and case history available	Integrated acceptance and failure-state checks

6. Technical Approach and IRS Mapping

6.1 Technique Mapping

IRS technique group	Implementation and evaluation object
Decision automation	Deterministic priority rules, four dispositions and separate reshipment action
Resource optimisation	Capacity-constrained VRPTW with greedy and OR-Tools Routing Solver
Knowledge discovery and data mining	Shipment failure classification, facility-level candidate causes and SHAP

Cognitive systems	Neo4j, template Cypher queries and evidence tracing

M1 and M7 provide engineering support. ML scores do not enter the decision API; the rule engine is the sole semantic source of online disposition recommendations.

6.2 Rule-Based Disposition

Inputs are product category, excursion temperature, duration in minutes, MKT, packaging condition and transport stage. Stage is recorded but currently does not affect rule branches. Historical exceptions are not decision features. The four category prototypes are vaccine_2_8, frozen_m20, insulin_2_8 and mrna_ultracold; none is a validated brand-specific configuration.

Rules run in priority order, stopping at the first match: freezing risk, compromised packaging with overtemperature, severe duration/MKT excursion, ordinary duration/MKT excursion, near-limit conditions for retestable products, and the remaining cases. Outputs are release / quarantine / retest / scrap. The current policy sets reshipment_required to true for quarantine or scrap; this logistics default requires separate evaluation.

rules_config.json and the Python engine define behaviour, with a semantic port for offline use. Sandbox overrides apply only to the current request and are saved at closure. Known limitations include low-temperature excursions for non-freeze-sensitive categories, approximating tiered stability with single thresholds, and the unresolved quarantine/scrap policy for rule 4. Rules or labels will not be changed merely to increase gold agreement.

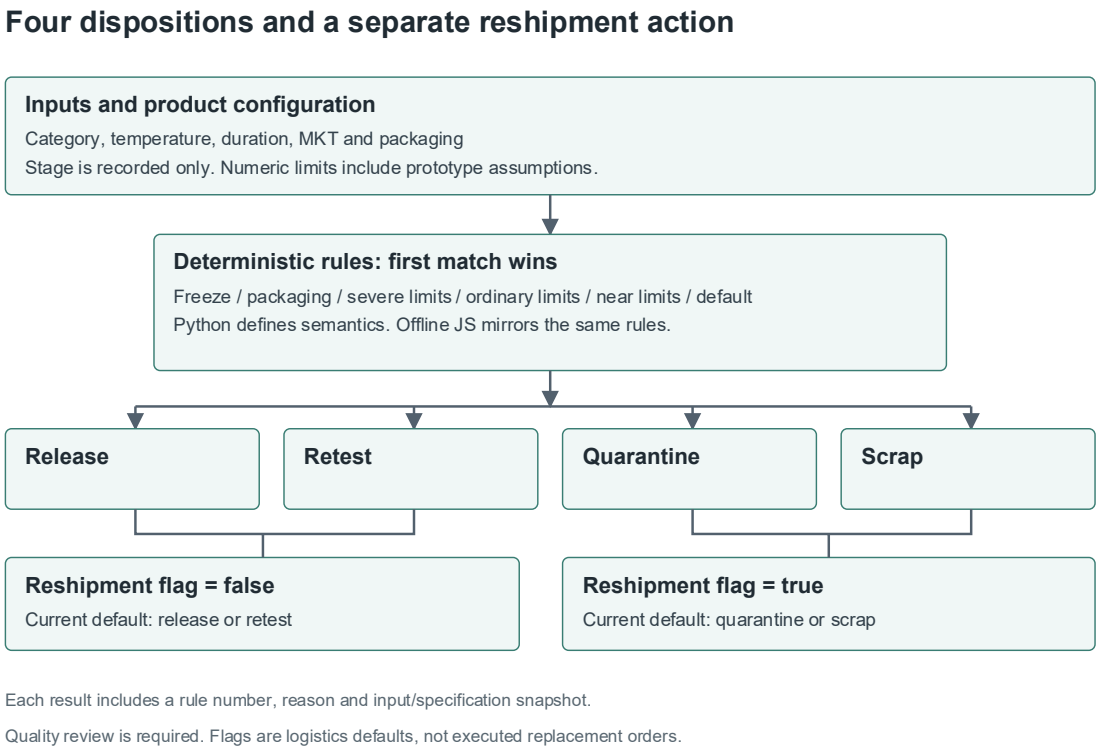


Figure 2: Four dispositions and an independent reshipment flag. Thresholds include prototype assumptions; recommendations require quality review.

6.3 Risk Classification and Candidate Causes

The risk task predicts silent_failure from 8,000 shipment summaries, comparing LR, LightGBM and XGBoost and contrasting interpretable features with an extended anonymous-feature set. The current split is 70/15/15, randomly stratified. The validation set selects the F1 threshold; imputation and scaling are fitted on training data only. Features include transit duration, temperature summaries, door openings and humidity statistics, not the current API's MKT or stage fields.

The dataset has no timestamp or forward-looking label relative to a prediction time. This is shipment-level failure classification, not demonstrated 30/60-minute early warning. The UI risk index and cause_code are deterministic heuristics, not ML inference.

Candidate-cause classification uses 16,192 facility-month simulation records with heat or freeze excursions and ten cause classes, currently split 80/20 with random stratification. Inputs cover equipment, power, monitoring and vaccine context. Their availability at diagnosis and repeated facilities across partitions require further checking. The legitimacy of outcome columns as inputs depends on the task's observation time. SHAP describes model associations, not proof of equipment-failure causation.

6.4 Delivery Optimisation

The model is a single-depot VRPTW with equal vehicle capacities, customer service windows, service times and depot-return limits, minimising travel distance. The implementation uses RoutingModel, PATH_CHEAPEST_ARC and GUIDED_LOCAL_SEARCH, not CP-SAT. A time-bounded solution has no global-optimality guarantee. A feasible greedy solution is a comparison baseline, not a lower bound. OR-Tools documentation (https://developers.google.com/optimization/routing/routing_options)

Six Solomon instances support standard comparisons. The Singapore case uses one depot, ten hospitals and real directed OSM matrices, measured in km and min. Travel times estimate free flow and exclude live congestion. Demands, fleet and service windows are simulated and do not imply actual commercial relationships among the facilities.

Vue shows precomputed routes for demonstration cases and fetches live routes for archived reshipment cases; temperature edits do not trigger solving. Single-depot reshipment solving is implemented over HTTP for an ordered set of destinations, quantities and windows. Alternative-stock selection, multiple temperature compartments, carbon and wastage-loss objectives are extensions. A genetic algorithm may be considered after integrated acceptance, but is not a mandatory deliverable.

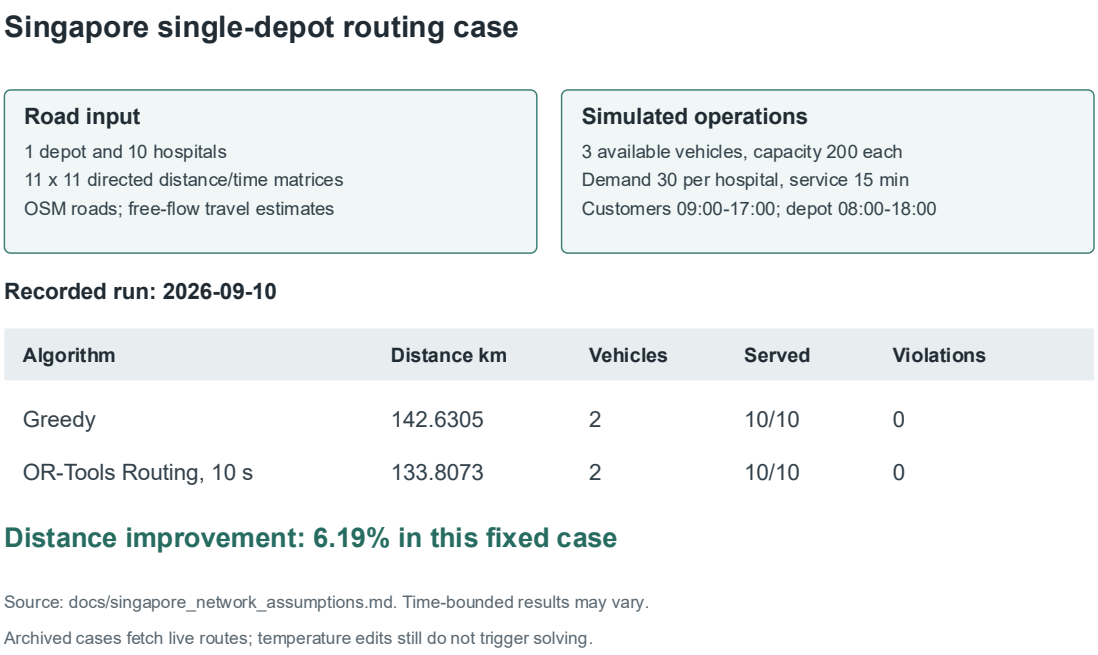


Figure 3: Real roads, simulated operations and recorded time-bounded results, not live vehicle tracking.

6.5 Knowledge Graph and QA

Neo4j entities include Product, Regulation, SOP, ExcursionEvent, Disposition, Cause, Facility, Shipment and ReshipmentOrder. Regulations, public procedural guidance, product configurations, Singapore facilities and dataset records retain source semantics. The approximately 8,000 Kaggle shipment records have unconfirmed real-world provenance and are suspected synthetic; they are not described as verified pharmaceutical shipments. Anonymous zones cannot be directly mapped to Singapore hospitals.

case_close records a case and attempts a graph write; decide previews are not archived. Cause currently represents the API heuristic code, not an ML prediction or investigated root cause. A reshipment node records an order; whether that order was solved and executed is tracked by dispatch run state, not by the graph node. Log and graph persistence are best-effort and do not provide validated audit-grade durability.

Python query functions support run_id-based explanations, audit chains, product requirements and statistics. /api/qa is implemented as bounded intent recognition over parameterised Cypher and returns case-specific evidence, distinguishing missing cases, insufficient evidence, unsupported questions and database failure; this is structured question-type routing, not natural-language classification. The frontend calls it in API mode and falls back to a keyword template offline. Evidence must support the particular rule, rather than merely provide a regulatory link. An LLM is not required for the MVP.

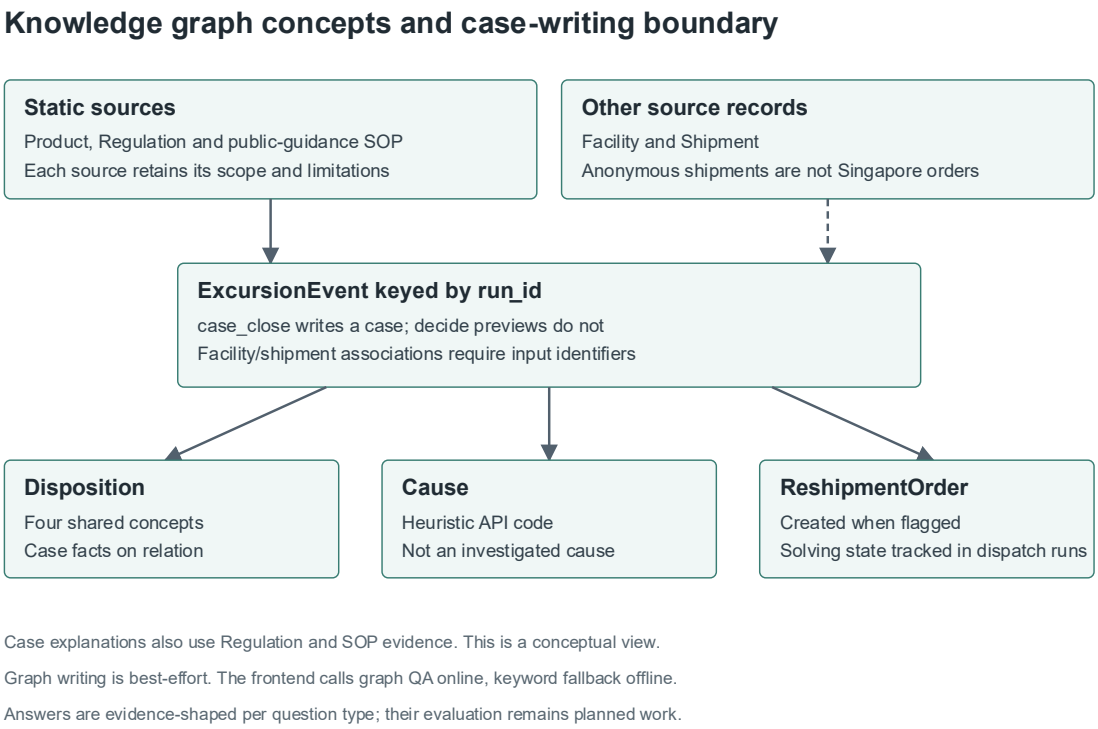
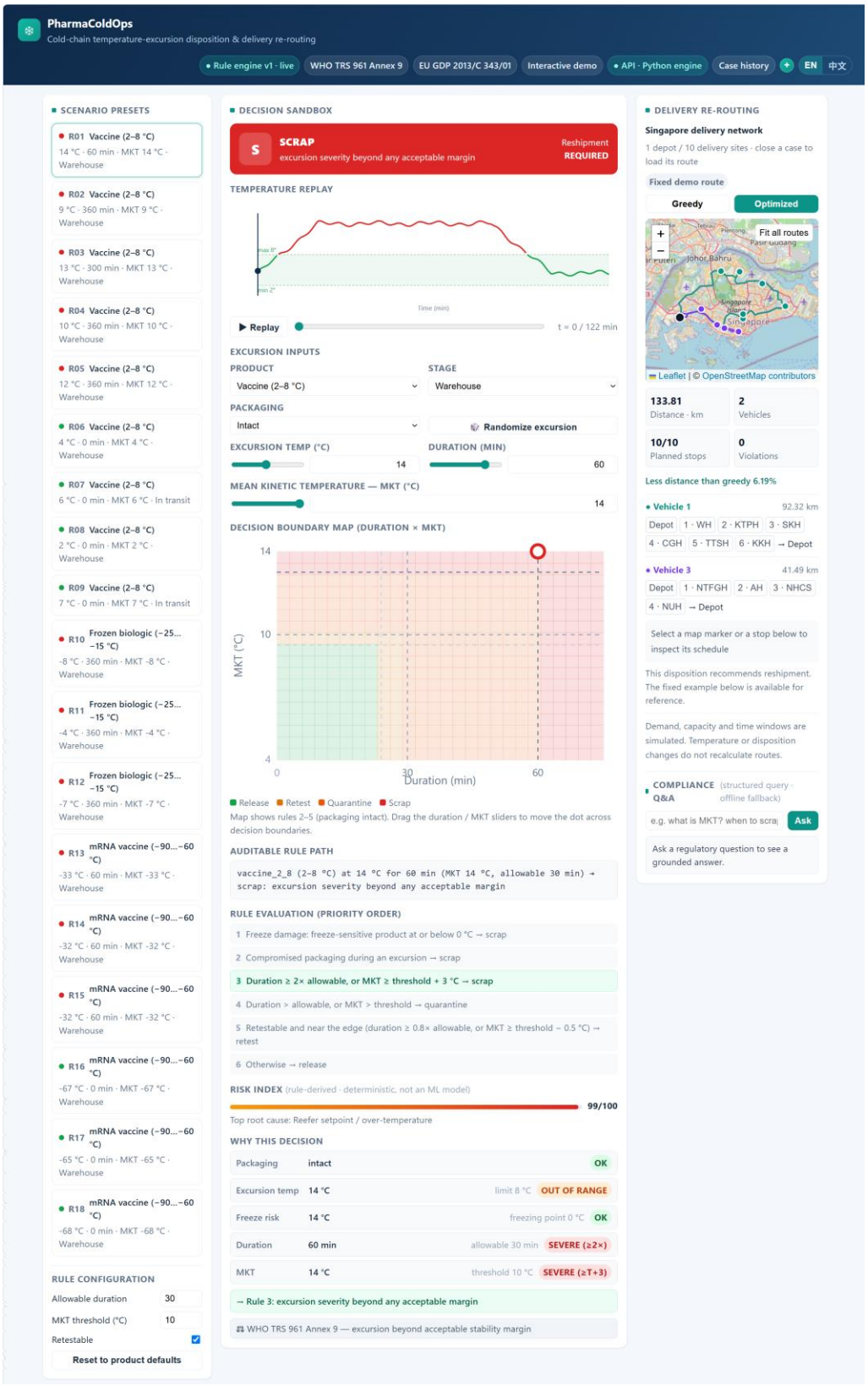
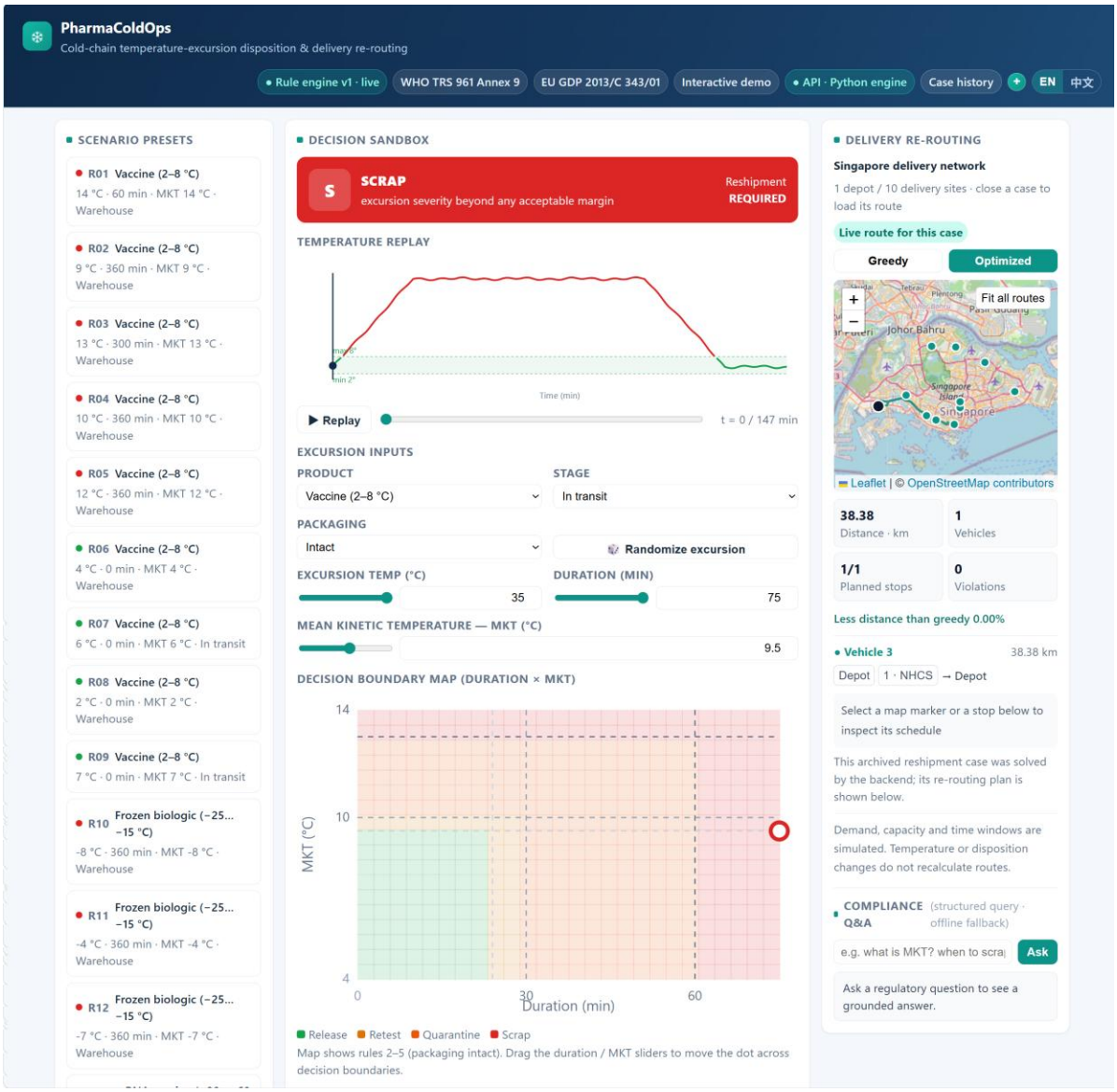


Figure 4: Conceptual case chain. Facility, shipment and route associations depend on the input contracts.



Demo, 2026-09-12: Rule recommendations, paths and heuristic risk scores, not final human approvals.



Demo, 2026-09-13: Precomputed routes for the demonstration case and an archived case fetching a live route and a case-specific graph answer. The order-driven dispatch screen is not shown because it is not built; this is not evidence of a completed integrated workflow.

6.6 Architecture and Interfaces

The frontend uses Vue 3, Vite, Pinia and Leaflet, with a legacy static demo retained. Online mode calls the Python engine through FastAPI/Uvicorn; offline mode uses a JS port. JSONL stores case history and Neo4j stores graph knowledge. Training and route-export scripts run separately.

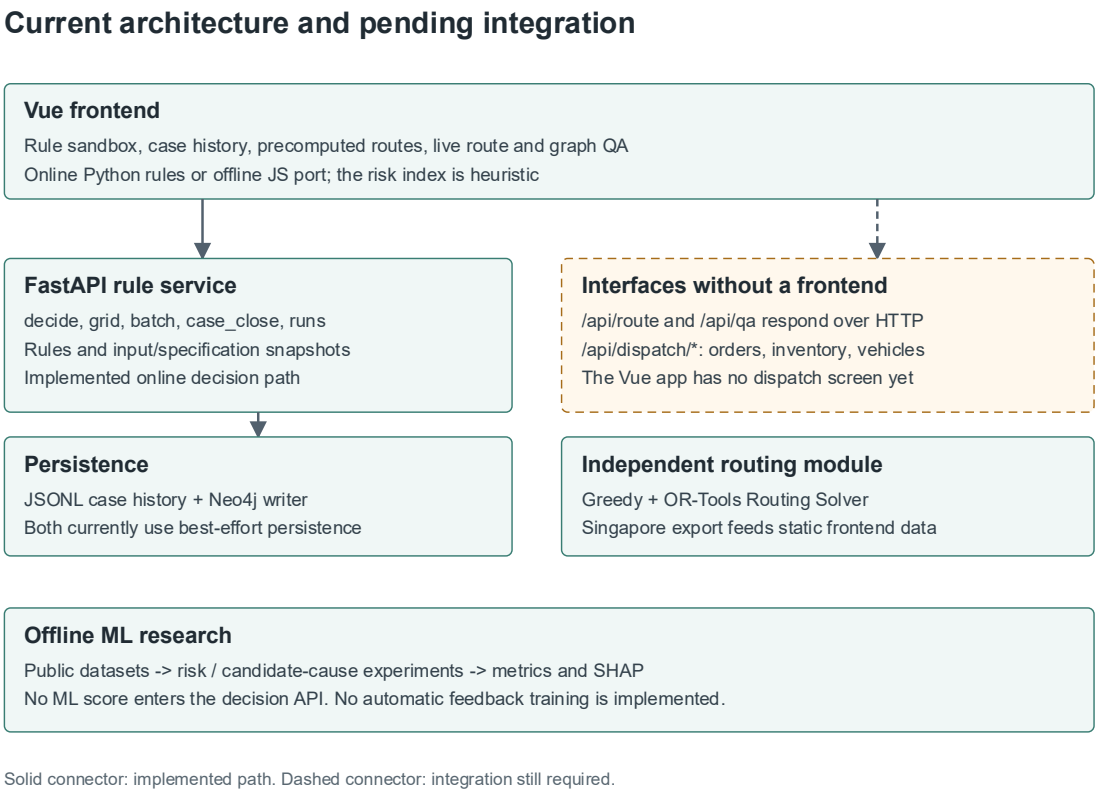


Figure 5: Online rules, offline experiments, precomputed routes and HTTP interfaces not yet wired to a screen are shown separately.

Cross-member contracts must fix the association among run_id, order_id, product, destination, quantity, time windows, temperature compatibility and ReplanResult, and handle infeasible routes, repeated closures and storage failures. The API now carries order destination, quantity, time windows, temperature zone and route results, with defined responses for infeasible routes and infeasible windows and a version-checked store for repeated commands; what remains on the contract is the frontend dispatch screen.

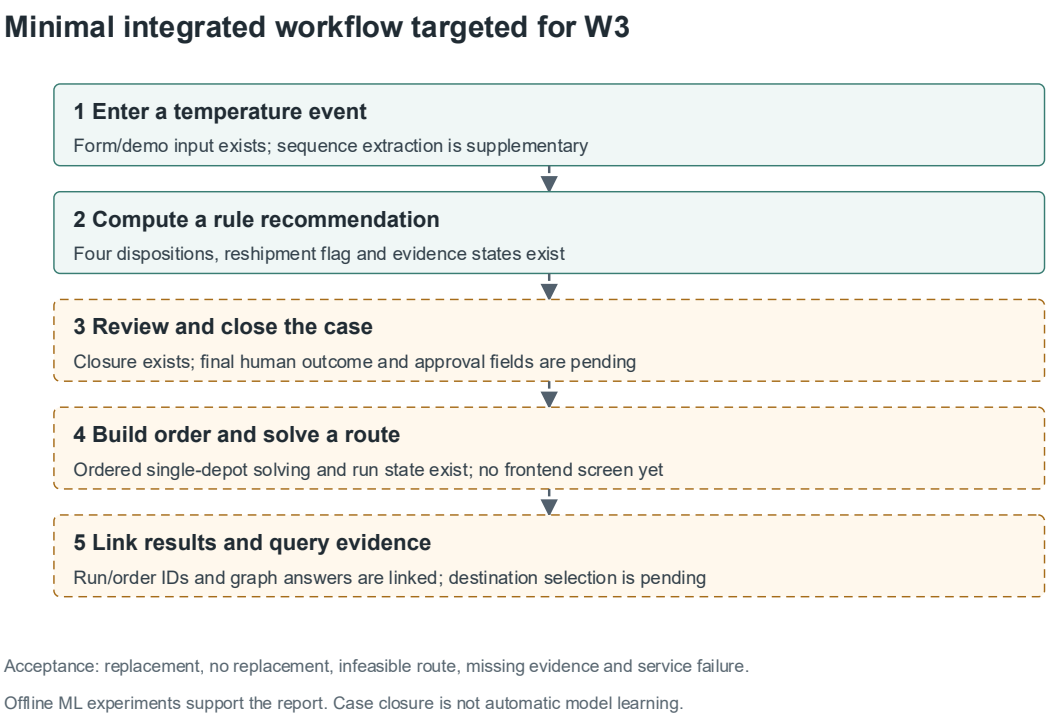


Figure 6: W3 integration acceptance target. Core modules exist, while connections remain to be implemented.

7. Data Collection and Preparation

7.1 Main Datasets

Dataset	Size and grain	Use	Nature and licence
Cold Chain Silent Failure (https://www.kaggle.com/datasets/skarin/cold-chain-shipment-silent-failure-dataset)	8,000 × 24; shipment summaries	Failure classification	Suspected synthetic; CC0
Vaccine Distribution (https://www.kaggle.com/datasets/manankhanna0/vaccine-distribution-with-temperature-logging)	26,674 × 13; logs for 30 batches	Demo material and sequence task	Suspected synthetic, real place names; CC0
Electric Sheep Africa vaccine-cold-chain (https://huggingface.co/datasets/electricsheepafrica/vaccine-cold-chain)	3 × 10,000 × 46; facility-month	Candidate causes	Simulation; CC BY 4.0
Africa Synth Immunization (https://huggingface.co/datasets/electricsheepafrica/africa-synth-immunization-vaccine-quality-cold-chain-all)	Approximately 30,000 rows	Optional quality-risk supplement	Synthetic; CC BY 4.0
Solomon / CVRPLIB (http://vrp.atd-lab.inf.puc-rio.br/index.php/en/)	Six committed instances	Standard VRPTW comparison	Abstract benchmark, preserve

			attribution
OpenStreetMap (https://www.openstreetmap.org/copyright) Singapore	11 facilities, 11×11 matrices, 110 directed paths	Local routing case	Real roads, simulated operations; ODbL
Authored scenarios and human gold	57 cases, four dispositions	Rule evaluation	Synthetic scenarios, independent annotation and arbitration

Licence verification is recorded in the data dictionary (../data/ml/DATA_DICTIONARY.md). Preserve Electric Sheep Africa attribution and licence information and the applicable OSM attribution. Other small-sample or cybersecurity datasets are not core benchmarks.

7.2 Domain Knowledge Sources

Rule evidence comes from WHO, EU GDP, ICH and applicable product documents. Procedural nodes cite public CDC/WHO guidance, not fictitious internal company SOPs. Record source, scope and assumptions, separating product evidence from general principles. Incomplete rule reviews are not described as fully verified.

7.3 Preparation and Task Boundaries

- Model each dataset at its original grain; do not invent common batch identifiers linking shipments, facility-months and Singapore orders.
- Use random stratification for the timestamp-free failure task. Multi-fold work must fit preprocessing and select thresholds inside each fold's training portion, reserving its test portion for scoring.
- Check repeated entities and feasibility of grouped or temporal facility-month splits, reporting differences from random splitting.
- Define product bands, windows, missing-data handling, MKT parameters and prediction time before the sequence task; prevent overlapping batch windows from leaking across partitions.
- Current demo export infers category from temperature, approximates MKT with one reading and fixes packaging as intact. These mappings cannot validate product stability or an MKT algorithm.
- Threshold-derived labels check event extraction, not the independent domain validity of disposition rules using the same thresholds.
- OSMnx builds directed distances, times and matching path geometry from cached roads, with source fingerprints. Order and vehicle assumptions are recorded separately.

8. Experiments and Evaluation

8.1 Metrics and Acceptance

Module	Metrics and checks	Interpretation boundary
Risk classification	F1, ROC-AUC, PR-AUC, precision/recall, multi-fold mean	No undefined early-warning metric

	and standard deviation	
Disposition rules	Gold agreement, macro-F1, class recall, confusion matrix, κ and rule coverage	Separate implementation fidelity, human agreement and domain validity
Reshipment	Policy agreement, order creation and duplicate triggering	Evaluate separately from the four dispositions
Candidate causes	Top-1, Top-3, macro-F1, support and failures	Simulation classification is not causal diagnosis
Routing	Distance, vehicles, summed duration, on-time rate, unserved orders, violations and runtime	Same instance, constraints and fixed budget
Graph QA	Correctness, case-relevant evidence coverage, no-evidence refusal and latency	Citation count is not relevance
System	End-to-end latency, identifiers, snapshot completeness and recovery	Separate preview, solving and closure timing

Routing reports unserved orders alongside on-time rates to avoid hiding failures. Published Solomon solutions require aligned fleet objectives, distance precision and constraints; one Singapore result does not establish operational savings. System acceptance covers reshipment, no reshipment, infeasibility, missing evidence, service failure and repeated closure.

8.2 Baselines and Preliminary Results

The following risk results were reproduced on 2026-09-13 by scripts/train_risk_full.py in the repository environment recorded in requirements.txt (seed 42). The data are suspected synthetic and the results describe that dataset, not real-world effectiveness. These are single-split results, not completed multi-fold evaluation.

Risk model, interpretable features	Validation threshold	F1	ROC-AUC	PR-AUC
Logistic regression	0.55	0.626	0.891	0.651
LightGBM	0.50	0.691	0.925	0.754
XGBoost	0.45	0.685	0.922	0.731

For causes, LightGBM recorded Top-1=0.143, Top-3=0.421 and macro-F1=0.125. The majority baseline recorded Top-1=0.161, Top-3=0.284 and macro-F1=0.028. Top-1 did not exceed the majority baseline, supporting further work on candidate ranking and features rather than a claim of reliable diagnosis. Formal evaluation must define an unambiguous Top-3 baseline and tie handling.

Recorded Singapore results from 2026-09-10:

Algorithm	Vehicles used	Customers served	Distance km	Summed route duration min	Violations / unserved

Greedy	2	10	142.6305	364.6972	0 / 0
OR-Tools Routing, 10 seconds	2	10	133.8073	357.5731	0 / 0

Distance decreased by approximately 6.19%. The case offers three vehicles of capacity 200 units each, demand of 30 units per hospital, customer windows 09:00–17:00 with 15-minute service, and depot hours 08:00–18:00. Duration sums vehicle working times. Time-bounded results may vary by environment. Experiment record (../PROGRESS.md), Singapore assumptions (../docs/singapore_network_assumptions.md)

8.3 Disposition Ground Truth

B and C independently labelled all 57 synthetic cases using only the written rubric, followed by A's arbitration. Human gold replaced engine-generated placeholder labels on 2026-09-12. Annotators are team members, not represented as domain experts. Individual responses remain in a private annotation package; the repository preserves arbitrated labels, findings and fingerprints.

Measure	Current result
Original B/C agreement	44/57
Inter-annotator Cohen's κ	0.6434, below the original 0.80 target
Current engine agreement with gold	39/57, 68.4%
Eighteen differences	15 engine quarantine/gold scrap; 2 engine release/gold quarantine; 1 engine retest/gold quarantine
Gold support	Release 13, retest 8, quarantine 3, scrap 33
High-consequence class recall	Quarantine 0/3; scrap 18/33, 54.5%

The team decided to report κ=0.6434 without revising and repeating the rubric exercise merely to increase κ, and to retain the three gold quarantine cases. Rule 4's quarantine/scrap choice remains an unresolved domain-policy question requiring product evidence. With only three gold quarantine cases, report support alongside metrics and avoid stable population-performance claims.

There are three evaluation levels: unit checks establish fidelity to frozen rules; human gold measures differences between written rules and team judgement; domain validity requires applicable product documents or external review. Tests freeze the eighteen known differences. Passing tests does not mean 100% gold agreement or prove all differences should be resolved by changing the engine.

This revision directly checked the real engine's 39/57 agreement and confusion counts. The selftest mode of evaluate_engine.py substitutes rubric outputs for an engine and validates the evaluation script only. Formal evaluation still needs a complete macro-F1, per-class precision/recall, κ and coverage table with version information. Annotation statistics (../data/processed/agreement_stats.md), Findings (../docs/annotation_findings_v1.md), Engine checks (../tests/test_rule_engine.py)

External product-level review is optional and has not been completed. Evidence-driven rule revisions should be versioned and assessed on independent new cases; tuning to the same gold set is not fresh independent validation.

9. MVP and Scope Control

Required: four explicitly prototype category configurations; four dispositions and separate reshipment semantics; single-depot simulated Singapore deliveries; rule-to-order-to-route integration; graph queries connected to the frontend; linked case records; independent rule evaluation and offline risk/cause experiments; bilingual UI and reproduction instructions.

Supplementary targets: sequence event generation and MKT validation, additional routing cases and stricter data splits. Prioritise the required workflow if time is constrained, disclose approximate demo inputs, and do not count unfinished supplementary work as delivered.

Optional extensions: early warning, tiered stability windows, multi-depot inventory selection, multiple temperature compartments, genetic algorithms, live vehicle positions, LLM polishing and feedback learning. Integrated acceptance takes priority.

Excluded: production ERP/WMS, actual IoT deployment, clinical or formal regulatory validation, patient data, automatic authorisation of real pharmaceutical dispositions and actual transport execution.

10. Plan and Team Responsibilities

10.1 Responsibilities

Member	Primary work	Shared deliverable
A (Xu Wenzhe)	Rules, domain evidence, human evaluation and report	Reshipment input contract and proposal review
B (Zhu Jianyu)	Data, risk, candidate causes and experiments	Data-nature disclosures and reported metrics
C (Wang Lepeng)	VRPTW, roads and route evaluation	Reshipment mapping and route output contract
D (Shen Ziyi)	KG, QA, API, frontend and demonstration	Case association, evidence display and videos

The table states primary ownership, not strict isolation. Delivery involved cross-assistance: rule thresholds and annotation wording were reviewed by both A and B, and the dispatch contract and frontend wiring were modified by both C and D, so a single module may show commits from more than one member.

10.2 Revised Weekly Plan

Start from work completed by 2026-09-12 and retain the original deadlines. The approximately 40 person-day budget prioritises integration, verification and delivery; optional algorithms are not mandatory. This is the revised proposal plan. Historical tasks in the daily plan do not mean those activities have yet to start.

Week	Main tasks and owners	Acceptance deliverable
W0 9/3–13	All: consolidate rules, data, annotation, routes and KG, revise proposal, add team details and synchronise materials	Review and submission due 9/13; group name, member names and student IDs added

W1 9/14–20	A: evidence/gold evaluation; B: task spec/multi-fold risk; C: order mapping; D: QA API/evidence checks	Evaluation tables, task specification and reviewable interfaces/queries
W2 9/21–27	A/C/D: order, route and case contracts/wiring; B: cause and split comparisons; D: frontend graph answers	Single-depot reshipment solving, real QA and failure states
W3 9/28–10/4	All: integration; A/D: closure and human-review record boundaries	Minimal workflow v1 with replacement, no-replacement and failure cases
W4 10/5–11	A: report/audit template; B: experiments/failures; C: Solomon and Singapore comparisons; D: QA evaluation	Formal results, relevant evidence checks and report draft
W5 10/12–18	All: integration tests and clean-environment reproduction; D: videos; A: report	Revised report, manual, video material and rehearsal
W6 10/19–25	All: final review, videos, packaging, reflections and submission	Final delivery by 10/25

Revised delivery schedule

Baseline: 2026-09-12. Original submission deadlines are retained.

Week	Dates	Acceptance milestone
W0	09/03-09/13	Proposal review, material alignment and submission
W1	09/14-09/20	Task specifications, rule evaluation and interface review
W2	09/21-09/27	Reshipment solving and frontend graph queries
W3	09/28-10/04	Minimal integrated workflow and failure cases
W4	10/05-10/11	Formal experiments, QA evaluation and report draft
W5	10/12-10/18	Integration checks, reproduction and video rehearsal
W6	10/19-10/25	Final review, videos, packaging and submission

Proposal due: 2026-09-13. Final delivery due: 2026-10-25.

W0 is not yet submitted. Optional algorithms follow integrated acceptance.

Figure 7: Deadlines retained, with remaining work focused on integration and evaluation.

11. Risks and Mitigation

Risk	Mitigation and disclosure
Engineering thresholds mistaken for stability limits	Identify evidence level, product scope and assumptions; retain human review
Annotation disagreement and unresolved policy	Report κ, confusion and small-sample limits; revise based on domain evidence
Synthetic data and mismatched grain undermine generalisation	Define tasks per dataset and check observation time and grouped leakage
Weak cause Top-1	Retain the majority baseline and analyse rankings and available features
Standalone modules without business integration	A/C/D prioritise order, destination, quantity, windows and case association
Irrelevant or missing QA evidence	Evaluate relevance per case and distinguish absent records, insufficient evidence and failure
Log or KG write failure	Test failure reporting, duplicate triggering and association completeness; no audit-grade durability claim
Infeasible or incomparable routing	Report unserved/violations and align constraints, objectives, units and budget
Scope and environment threaten delivery	Defer extensions, record dependencies/reproduction and retain offline demonstration
Data use and provenance	Preserve licences, attribution and fingerprints; keep personal annotations private

12. Final Deliverables

- GitHub repository containing rules, data preparation, model experiments, optimisation, KG/API and frontend.
- Formal report, user manual, personal reflections and a submission archive organised to course requirements.
- Two five-minute videos covering the workflow and methods/evaluation, identifying simulated and incomplete parts.
- Reproduction instructions, rule assumptions and sources, annotation findings, route metrics and QA evaluation cases.

The Markdown files are the revised bilingual proposal sources, and the Chinese Word proposal has been regenerated from this version. Before submission, check PPT consistency and complete the submission records. Content revision does not itself mean W0 has been submitted.

13. Conclusion

PharmaColdOps now has a rule-decision core, independent human annotations, offline experiments and routing over real Singapore roads. Results also expose specification disagreements, limited cause-classification performance and gaps in data provenance and product-threshold evidence, informing the remaining evaluation.

The remaining priority is to connect replacement demand, route results and case evidence into an explainable, reproducible workflow with explicit applicability limits by 2026-10-25. Integration and evaluation demonstrate the four IRS technique groups; unverified clinical, operational or compliance benefits are not completion criteria.

14. References and Project Evidence

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- Google. OR-Tools Routing Options (https://developers.google.com/optimization/routing/routing_options).
- OpenStreetMap contributors. Copyright and ODbL (https://www.openstreetmap.org/copyright).
- Dataset links appear in §7.1. Project evidence: DATA_DICTIONARY (../data/ml/DATA_DICTIONARY.md), PROGRESS (../PROGRESS.md), Network assumptions (../docs/singapore_network_assumptions.md), Annotation statistics (../data/processed/agreement_stats.md), PROVENANCE (../data/processed/PROVENANCE.md).
- NUS-ISS. IRS proposal and final presentation guidelines v016 (IRS%20practice%20module%20project%20proposal%20%26%20final%20presentation%20guidelines%20v016.pdf), course-supplied material.

15. Supplementary Note on AI Use

Generative AI assisted drafting, translation, layout and cross-checking: Claude for earlier drafts and Codex for this revision. AI also assisted scenario drafts and development. Human gold comes from independent B/C

annotation and A's arbitration. AI output is not stability or compliance evidence. The team is responsible for citations, rule assumptions, experiment reproduction and submission materials. Incomplete product-level and external validation are disclosed in the proposal.